Dual-Branch Network for Spatial–Channel Stream Modeling Based on the State-Space Model for Remote Sensing Image Segmentation

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***Abstract*— To quickly and effectively address the color similarity issue in remote sensing image segmentation, traditional channel attention methods typically use channel modeling based on global channel statistics mapped to weights. However, this approach either suffers from limitations in feature selection due to a lack of dynamic interactions and the loss of significant spatial information, resulting in poor performance in complex scenarios, or has high computational complexity, making it difficult to apply in high-resolution remote sensing images. To overcome these challenges, this article proposes an innovative streaming channel modeling method based on state-space models (SSMs), aimed at rapidly and efficiently tackling the color similarity problem in remote sensing image segmentation. Specifically, we designed the channel-position state-space model network (CPSSNet) framework, where the decoder comprises the spatial Mamba block (SMB) for spatial modeling and the channel Mamba block (CMB) for streaming channel modeling. The core component of SMB, position-selective-scan-2D, achieves multidirectional global modeling in the spatial domain through a combination of the spatial scanning algorithm and SSM, with linear complexity. The core component of CMB, channel-2D selective-scan (C-SS2D), fuses channel and spatial information into patches for streaming modeling using a combination of the channel scanning (CS) algorithm and SSM. We have further improved SSM within C-SS2D to enhance dynamic interactions between channels, allowing for more refined modeling while maintaining linear complexity. Experimental results demonstrate that CPSSNet exhibits outstanding performance in addressing color similarity challenges in remote sensing image segmentation. The code is available at https://github.com/yysdck/CPSSNet.**

***Index Terms*— Channel modeling, Mamba, remote sensing, semantic segmentation, state-space model (SSM).**

I. Introduction

With the advancements in sensor and space technology, acquiring high-resolution satellite and aerospace remote sensing images has become increasingly accessible. These images provide high-resolution data for observing diverse landscapes on the Earth's surface, including urban areas, farmlands, forests, and lakes. Remote sensing image segmentation, a key technology, aims to partition images into distinct land objects or categories, playing a crucial role in geographic information systems (GISs), resource management, environmental monitoring, and crisis management. Its key applications include land cover mapping [1], [2], change detection [3], [4], environmental protection [5], [6], road and building extraction [7], [8], and other practical applications [9], [10].

In the field of remote sensing segmentation, various deep learning-based methods have been proposed, covering hyperspectral images (HSIs), optical images, lightlaser detection and ranging (LiDAR) data, and integrated multisensor data [11], [12]. Convolutional neural networks (CNNs) have shown significant success in classifying each pixel in an image with semantic labels [13]. Many researchers continue to explore CNN-based segmentation networks, such as AFF-Unet [14] and CATNet [15].

Although CNNs perform excellently in remote sensing image segmentation, they still face challenges when dealing with complex land cover characteristics [16]. First, the fixed-weight mechanism of convolutional layers cannot dynamically adjust the contribution of features, making it difficult to optimize the expansion and contraction of features based on specific contexts. Second, the local pattern design of convolutional layers limits the network's ability to capture spatial long-range dependencies, which significantly affects the recognition of complex land cover features.

To address the issue that convolution cannot dynamically adjust the feature contribution, researchers have introduced various channel attention mechanisms, such as the squeeze-and-excitation module (SEM) in SENet [17], adaptive fine-grained channel attention (AFGCA) in UBRFCNet [18], and MECA [19], among others. These methods rely on global spatial statistics as weights, neglecting local details, which leads to the loss of spatial information and limits dynamic interactions between channels. Although serial [20] and parallel [20], [21] methods combining spatial and channel attention have made some progress, they still struggle to fully recover the lost spatial information. While self-attention methods [22] have improved the flexibility of channel weighting and can utilize more spatial information, they typically have higher computational complexity. Therefore, achieving low-complexity dynamic channel feature interaction to quickly and effectively address color similarity issues (seeing Fig. 1) in remote sensing image segmentation remains a key challenge that needs to be solved.

Received 4 December 2024; revised 16 January 2025; accepted 10 February 2025. Date of publication 24 February 2025; date of current version 7 March 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 61772319, Grant 62272281, Grant 62002200, and Grant 62202268; in part by Shandong Natural Science Foundation of China under Grant ZR2023MF026 and Grant ZR2024QF258; and in part by Yantai Science and Technology Innovation Development Plan under Grant 2023JCYJ040. *(Corresponding author: Genji Yuan.)*

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Digital Object Identifier 10.1109/TGRS.2025.3544736

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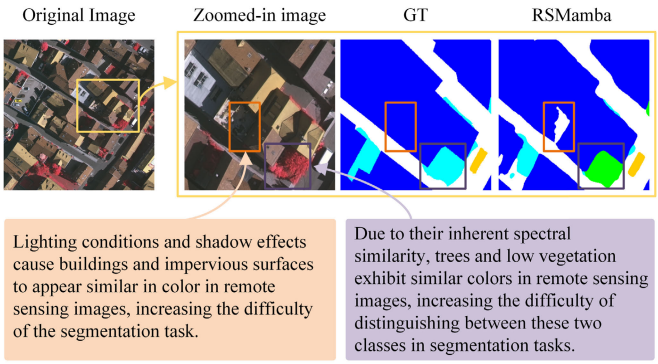


Fig. 1. Challenges posed to segmentation accuracy by the issue of color similarity in remote sensing image segmentation, caused by similar lighting conditions or highly similar spectral features of the objects.

To overcome the local perceptual limitations of convolution, the self-attention mechanism of Transformers has become a mainstream method for handling spatial long-range dependencies in remote sensing image segmentation, as seen in Segmenter [23], EMRT [24], and others [25]. However, these methods tend to have high computational complexity in dense prediction tasks for remote sensing images, especially in high-resolution image processing [26], [27].

Recently, general Transformer-based segmentation models, such as SAM [28] and RSPrompter [29], have made some progress, but they usually rely on prior knowledge (e.g., points, boxes, or masks), and their computational complexity remains high, making them less practical in real-world applications. Therefore, developing more efficient alternatives has become especially important. Although various efficient attention mechanisms [30], [31], [32] have been proposed to reduce computational overhead, it remains difficult to achieve an ideal balance between parameter dependency, computational speed, and expressive power.

To break through the limitations of traditional long-range dependency methods that rely on high-complexity self-attention mechanisms, the latest work, Mamba, integrates time-varying parameters into state-space models (SSMs) [33], introducing hardware-aware algorithms to improve training and inference efficiency. Inspired by Kalman filtering [34], SSM can implicitly capture long-range dependencies and leverage the advantages of parallel training to achieve linear complexity for long-range modeling, making it a potential alternative to self-attention in language modeling [33]. However, Mamba is designed for directional data and is not suitable for images. To address this, Vim [35] implements selective scanning along the horizontal axis, and VMamba [36] employs this strategy in both horizontal and vertical directions to enhance the effective receptive field.

In the field of remote sensing images, models, such as PyramidMamba [37], RS-Mamba [38], RS^3-Mamba [39], and Samba [40], have established remote dependencies with linear complexity in remote sensing images using the Mamba backbone. However, these SSM-based methods independently apply at the channel level, neglecting cross-channel information flow (i.e., channel mixing), which leads to a loss of relationships between features and fails to dynamically adjust feature contributions. This limitation restricts the model's ability to capture complex feature interactions and weakens its sensitivity to subtle feature variations in remote sensing images.

Inspired by the aforementioned studies, this article proposes channel-position state-space model network (CPSSNet), a network designed for remote sensing segmentation tasks. It leverages the SSM to establish long-range dependencies on the spatial level while focusing on channel information stream modeling to enable dynamic feature interaction, all while maintaining linear complexity. CPSSNet consists of three components: an encoder, a patch fusion module, and a decoder. The encoder performs conventional feature extraction, followed by patching after multiscale feature fusion. The decoder includes two parallel visual SSM blocks—spatial Mamba block (SMB) and channel Mamba block (CMB). SMB, based on the VSS block of VMamba, employs the position-selective-scan-2D (P-SS2D) core component with a selective scanning (SS) algorithm to achieve directional spatial scanning and establish spatial long-range dependencies. CMB's core component, channel-2D selective-scan (C-SS2D), utilizes a channel scanning (CS) algorithm to dynamically model channel dependencies, implicitly capturing relationships between adjacent features through weighted SSM, also with linear complexity. Past channel modeling methods assign a weight to each channel, which lacks fine-grained dynamic channel interactions and results in the loss of significant spatial information. As a result, traditional methods have limitations when handling land cover with high color similarity, struggling to effectively distinguish between categories with similar colors or grayscale values, leading to misclassification and blurred boundaries, especially in remote sensing images. For example, green regions in forests and farmlands may be misclassified. Our dynamic channel streaming modeling method overcomes this issue by introducing learnable weights dependent on the previous state, enabling the model to adaptively derive channel weights during inference. This approach better addresses the challenges posed by color similarity, significantly improving segmentation accuracy and offering greater scalability.

In summary, the contributions of this article are as follows.

1) We propose the CMB module, where the core component, C-SS2D, first serializes two channel planes in four directions via CS and then models them using a designed weighted SSM, establishing patch-level implicit channel dependencies with linear complexity. This is a novel exploration of SSM in channel modeling.
2) Based on the channel-level CMB and spatial-level SMB, we propose CPSSNet, which enables modeling of long-range dependencies at the spatial level and streamlining of channel information with linear complexity. This architecture overcomes the traditional issues of losing significant spatial information and channel modeling that does not rely on interactive information weights. It realizes dynamic channel information flow, enhancing the model's focus on dynamic interactions between adjacent features.

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3) In multispectral remote sensing images, due to the spectral features of certain objects being highly similar in specific bands, models based on lack of interactive channel weighting and insufficient spatial information tend to ignore the dynamic interaction of rapidly changing and overlapping features, thereby exacerbating the issue of color similarity. Our CPSSNet effectively addresses this problem through a streaming modeling approach, achieving state-of-the-art results on three mainstream benchmark datasets: ISPRS Vaihingen, ISPRS Potsdam, and LoveDA.

II. Related Work

Remote sensing image segmentation is a crucial task aimed at partitioning images into different regions based on land cover categories or features. In this field, researchers are continuously striving to improve the accuracy and efficiency of segmentation results, exploring various methods and techniques. Successful remote sensing semantic segmentation not only has important applications in urban planning, agricultural monitoring, and other fields but also plays a crucial role in environmental monitoring and resource management. This section will introduce some important work related to remote sensing segmentation and analyze the impact and contributions of these works to the field.

A. Semantic Segmentation of Remote Sensing Images Based on Attention Mechanism

Convolutions have certain limitations. First, the local pattern design of convolutional layers restricts the network's ability to capture long-range spatial dependencies. Second, the fixed-weight mechanism of convolutional layers cannot dynamically adjust feature contributions during feature selection, preventing the expansion and contraction of features from being optimized based on specific contexts.

Researchers have proposed various solutions to address the local pattern limitations of CNNs. In remote sensing image segmentation, attention mechanisms are particularly important as they enhance the model's understanding of global context. Self-attention mechanisms, such as ViT and Swin Transformer, have become mainstream methods for achieving global spatial modeling. However, other attention mechanisms have also been proposed for remote sensing segmentation, such as the linear attention mechanism by Li et al. [41], the local attention block by Ding et al. [42], and the local channel–spatial attention and multiscale attention in MSCSA-Net [43]. Despite this, these methods still rely too heavily on local features extracted by CNNs and fail to directly encode global information. The self-attention mechanism can effectively model long-range dependencies, which has led to its widespread development in the field of image [23], [44], [45].

In the domain of large models, general segmentation models such as SAM [28] and remote sensing segmentation models such as RSPrompter [29] have also made significant progress. However, these large models often rely on prior knowledge (such as points, boxes, or masks), and the self-attention mechanism requires explicit maintenance of the attention matrix, introducing quadratic complexity. This results in limited practical application in high-resolution remote sensing images. To address this, researchers have proposed various efficient attention mechanisms [30], [31], [32], [46], [47], [48] aimed at reducing computational complexity and sparsifying the attention matrix. However, these methods often involve a tradeoff between expressiveness and speed, leading to insufficient scalability and accuracy, and lower efficiency in practical applications.

Channel attention mechanisms, such as SENet [17], MECA [19], and UBRFCNet [18], perform weighting by compressing each channel into fixed statistics. This approach leads to a significant loss of spatial information, making it ineffective in enhancing feature representation capability, and lacks dynamic interaction between channels. In multispectral remote sensing images, the spectral features of certain land cover types are very similar in specific bands, causing them to nearly overlap in the image. This makes it difficult for models lacking dynamic interactions and spatial information to effectively distinguish between these similar land cover classes, and the issue of color similarity caused by feature overlap and rapid changes remains inadequately addressed.

Although methods combining spatial attention, such as BAM [21], CSAB [20], and ReSCNet [20], attempt to address this limitation, the irreversible nature of channel attention still results in information loss that cannot be fully compensated for. Self-attention methods such as DANet [49] and TCA [50] have achieved fine-grained channel dependency modeling, but they still require maintaining attention matrices, resulting in quadratic complexity, and they struggle to address the issue of color similarity in multispectral remote sensing images.

In contrast, this article employs an SSM to achieve spatial dependencies with linear complexity while fully considering the dynamic streaming channel modeling of spatial information. Through this dynamic modeling, CPSSNet is able to more effectively handle feature overlap and rapid changes in multispectral remote sensing images, significantly improving its ability to address color similarity issues and ultimately enhancing segmentation accuracy.

B. State-Space Models

Recently, significant progress has been made in the field of deep learning with the introduction of SSMs, bringing about major changes in modeling remote dependencies and sequential data processing methods. Subsequently, the integration of SSMs with deep learning has been significantly enhanced through the introduction of the HiPPO initialization method, which has improved the model's ability to capture remote dependency relationships [51].

The release of the LSSL model marked a significant step forward in addressing long-term dependency challenges with SSMs [52]; however, the limitations in computational and memory efficiency of LSSL remain to be addressed. In response to this challenge, researchers have proposed the S4 model [53], which successfully reduces computational overhead by adopting a novel parameterization strategy, making SSMs more feasible for practical applications.

As the S4 model continues to improve, the field of SSM research has rapidly expanded, leading to the emergence of

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various variants to further enhance the structure and efficiency of the models. For instance, some models employ complex diagonal structures to enhance temporal modeling capabilities [54], while others support configurations with multiple inputs and outputs [55], further enhancing the applicability and flexibility of SSM models.

However, existing SSM models are limited to processing unidirectional sequence data and cannot effectively handle image data lacking specific orientation. To address this issue, recent studies have leveraged selective scanning methods [35], [36] to achieve global modeling of image contexts in specific directions. For example, Vim [35] adopts selective scanning in the horizontal direction, enabling each image region to perceive global information, while VMamba [36] further enhances the model's global perception range through selective scanning in both horizontal and vertical directions.

In the field of remote sensing image segmentation, SSMs have led to the emergence of several innovative methods due to their outstanding performance in modeling long-range dependencies, significantly broadening the research scope of remote sensing image analysis. As the first model to apply SSM to remote sensing dense prediction tasks, RS-Mamba [38] pioneered the validation of SSM's applicability in this domain, laying the theoretical and practical foundation for subsequent research. Following RS-Mamba, several backbone networks, such as PyramidMamba [37], changemamba [56], RS^3 Mamba [39], CDMamba [57], and Samba [40], have been proposed. By incorporating multilevel and multiscale feature fusion strategies, these models have significantly enhanced the spatial context modeling ability. At the same time, by optimizing computational processes and structural designs, they have successfully reduced computational complexity, excelling in efficiently handling complex remote sensing image segmentation and change detection tasks.

However, current research mainly focuses on applying SSM to spatial structure modeling, with limited exploration of its potential at the channel level. In remote sensing image segmentation, the channel dimension holds rich semantic information, essential for distinguishing features and handling complex scenes, especially when color changes are subtle and spectral information overlaps. Effective modeling and weighting of channel features are critical in these cases, improving segmentation accuracy and detail preservation.

To address this, we propose a method that extends SSM to the channel dimension by considering channel discreteness. By establishing channel dependencies and using a nonlinear weighting mechanism, we achieve deep fusion of channel and spatial information. This method significantly improves segmentation robustness and accuracy, particularly in complex background scenarios.

III. Method

CPSSNet is composed of three important parts: an encoder part for feature extraction, a fusion patching part for multiscale feature fusion and patchification, and a decoder part consisting of channel-level CMB and spatial-level SMB. We will subsequently provide a detailed introduction to our decoder and the C-SS2D method embedded within CMB. Finally, we will introduce the loss function we utilized.

A. CPSSNet Architecture

The structure of CPSSNet is illustrated in Fig. 2. First, ConvNext [58] is employed as the backbone network, which performs encoding across four stages to extract features at four different scales. Subsequently, in the fusion patching stage, the features from the last three stages are fused to obtain multiscale mixed features with stronger representational capability. Since the mixed features still have large dimensions, we apply a patch embedding operation, dividing the mixed features into smaller patches, each transformed into a fixed-length embedding tensor. In this stage, patching operations are conducted not only on the spatial dimension but also on the channel dimension. Specifically, for an input feature F_{in} with dimensions (C_{in}, H_{in}, and W_{in}), we first adjust the channel dimension to match the height and width, resulting in dimensions (C, H, and W), where $C = W = H$. Then, we reshape along the channel–height (C–H) and channel–width (C–W) directions to perform a patching operation using a 2-D convolution with a kernel size and stride equal to the patch size. This generates spatially correlated channel indicators. This stage outputs three matrices: a spatial tensor with dimensions (C, H_p, and W_p), and two channel tensors with dimensions (C_p, H_p, and W) and (C_p, H, and W_p), where C_p, H_p, and W_p denote the patched channel, height, and width, respectively. The corresponding relationships are given by

$$C_p, H_p, W_p = \frac{C}{\text{PS}}, \frac{H}{\text{PS}}, \frac{W}{\text{PS}} \tag{1}$$

where PS denotes the patch size.

Next, the patched tensors are input into the decoder. The decoder of CPSSNet consists of two parts: the SMB for spatial dimension and the CMB for channel dimension. The SMB uses SSM to establish spatial long-range dependencies within the original feature matrix, while CMB uses SSM to model the streaming channel information within the original feature matrix. Finally, the outputs of these two SSM blocks are combined and integrated with the features from the first stage of the original feature extraction and then fed into the segmentation head. This method adaptively adjusts the weights of spatial and channel features, enhancing the network's ability to capture important features in the image while attenuating irrelevant ones. This method adaptively adjusts the weights of spatial and channel features, enabling the network to better capture important features in the image while diminishing the influence of irrelevant features. Compared to similar approaches like UBRFCNet [18], this method considers more spatial information in its modeling. In contrast to C^{2}TNet [22], which employs self-attention at the channel level, our method achieves linear complexity. Fig. 3 shows different channel modeling methods. Furthermore, the aforementioned approach does not take into account the interaction between channel information, whereas our solution employs a dynamic flow-based modeling approach that relies on the previous hidden state, offering greater scalability.

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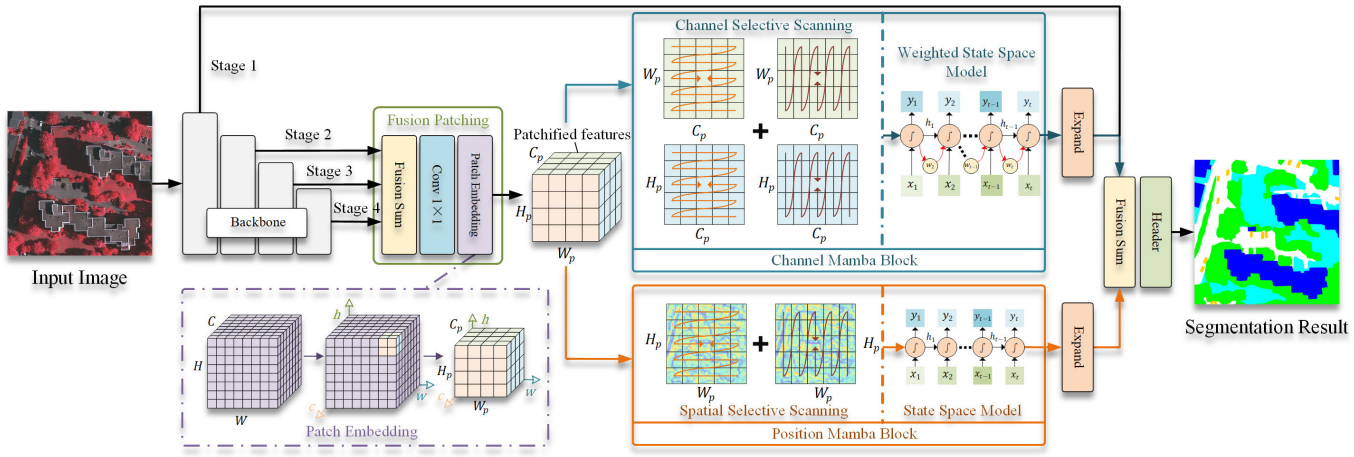


Fig. 2. Architecture of CPSSNet. The input image is first sampled to obtain multiple features at different scales, which are then fused and patchified. This patchification occurs at both the spatial and channel levels. Next, the patchified features are serialized in different directions and processed by CMB and SMB to establish channel and spatial dependencies, respectively. Finally, the outputs of CMB and SMB are fused and passed through the segmentation head to obtain the final segmentation map.

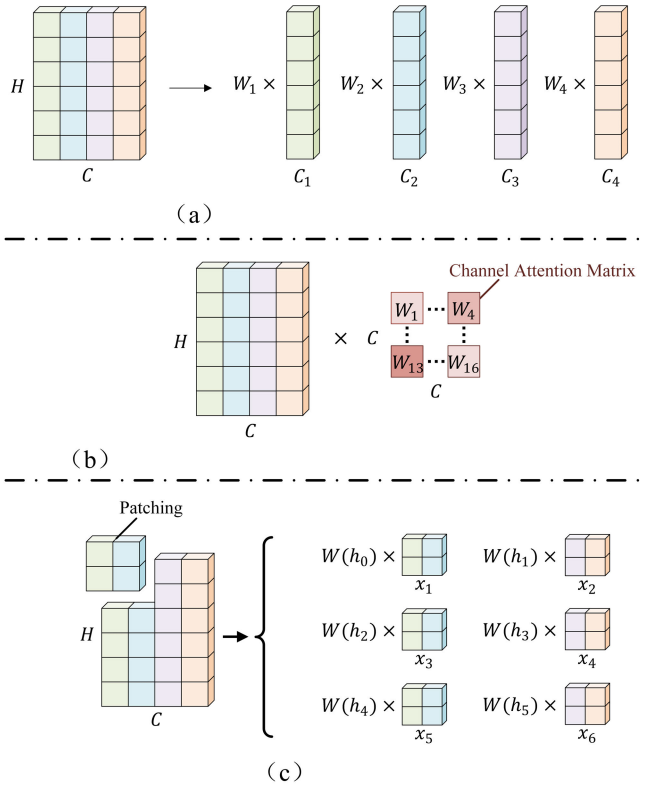


Fig. 3. Comparison of three different channel-weighting methods is shown. (a) Traditional channel attention method. (b) Channel attention method constructed by self-attention. (c) Our proposed C-SS2D weighting method uses a channel-flow modeling approach for weighting.

B. Decoder

In this section, we will introduce the implementation of the two SSMs in the decoder. Before that, we will first present the preliminary knowledge on SSM.

1) State-Space Model: The SSM is considered a linear time-invariant (LTI) system that maps an input sequence $x(t) \in \mathbb{R}$ to an output sequence $y(t) \in \mathbb{R}$ through a hidden state $h(t) \in \mathbb{C}^N$. These mappings are typically represented by linear ordinary differential equations (ODEs) as follows:

$$h'(t) = Ah(t) + Bx(t) \tag{2}$$

$$y(t) = Ch(t) + Dx(t). \tag{3}$$

Equation (2) is the state equation, and (3) is the output equation. Here, $A \in \mathbb{C}^{N\times N}$ is the state transition matrix describing how the state evolves over time, $B \in \mathbb{C}^N$ is the input control matrix describing how the input influences state evolution, $C \in \mathbb{C}^N$ is the observation matrix describing how the state maps to the output, and $D \in \mathbb{C}^1$ is the direct input matrix describing how the input directly affects the output. Given a time step $\Delta \in \mathbb{R}$, the continuous model is discretized using the zero-order hold (ZOH) method, resulting in

$$h_t = \bar{A}(h)_{t-1} + \bar{B}x_t \tag{4}$$

$$y_t = Ch_t + Dx_t. \tag{5}$$

Here

$$\bar{A} = f_{\bar{A}}(\Delta, A) = e^{\Delta A} \tag{6}$$

$$\bar{B} = f_{\bar{B}}(\Delta, A, B) = (\Delta A)^{-1}\left(e^{\Delta A} - I\right)\Delta B \tag{7}$$

where $f_{\bar{A}}$ and $f_{\bar{B}}$ are discretization functions, representing ZOH. For (7), the first-order Taylor series expansion gives

$$\bar{B} \approx \left(\Delta A^{-1}\right)(\Delta A)\Delta B = \Delta B. \tag{8}$$

Using (4)–(6) and (8), the output y can be derived and represented as the output sequence $y = [y_1, y_2, \ldots, y_t, \ldots, y_L]$.

2) Spatial Mamba Block: As the module for establishing dependencies in the image spatial domain, we incorporate the VSS block from VMamba [36]. The structure of SMB is shown in Fig. 4(a). For the patchified tensor, LayerNorm is first applied, followed by splitting into two branches. In the first branch, the tensor passes through a linear layer and then is activated using SiLU. In the second branch, the normalized tensor is processed by a linear layer, depthwise separable convolutions, and an activation function before being input into a P-SS2D module for establishing spatial long-range dependencies. Afterward, the output is element-wise multiplied with

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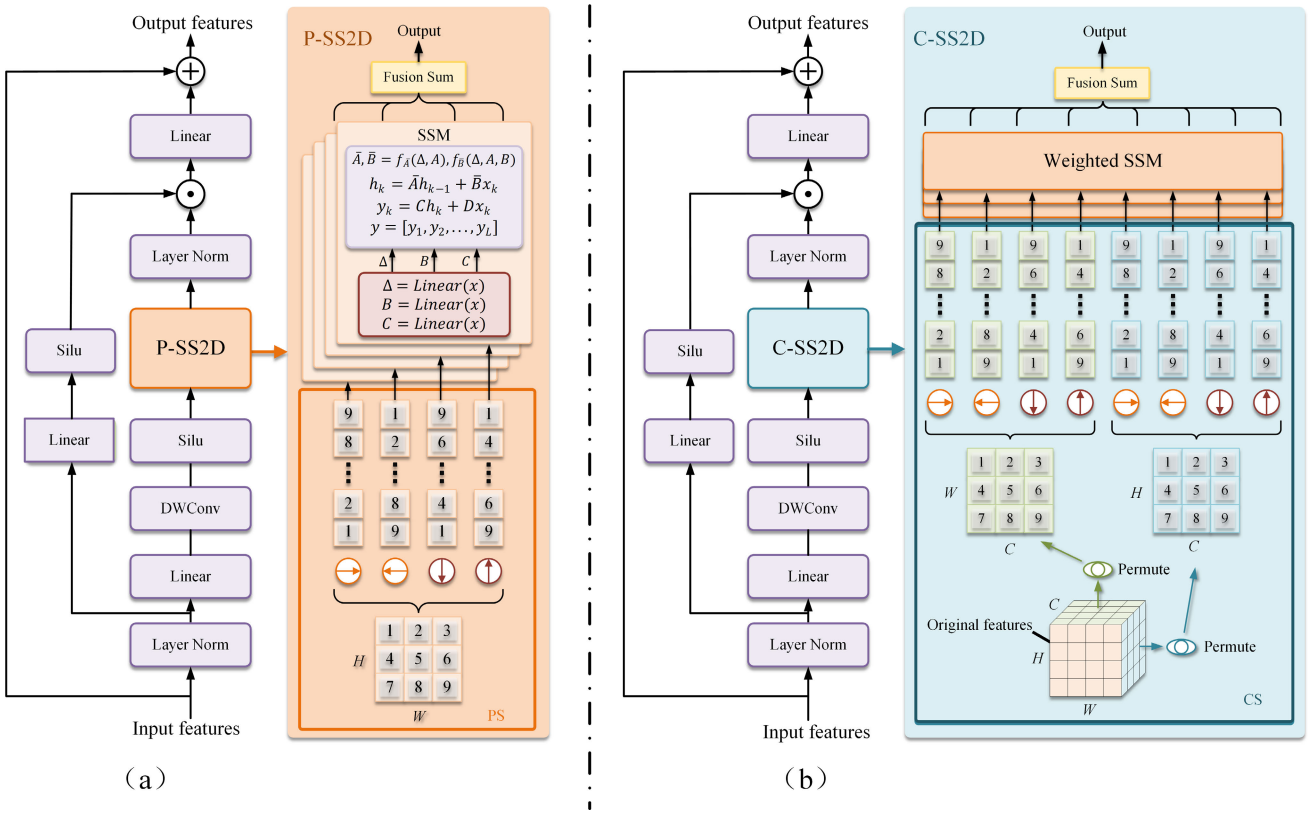


Fig. 4. Two key modules of the decoder. (a) SMB structure, with P-SS2D as its core component. (b) CMB structure, with C-SS2D as its core component.

the output of the first branch following layer normalization. Finally, the features are mixed through a linear layer and connected with the initial tensor via residual connection to produce the output of the SMB block.

P-SS2D, the core component of SMB, consists of three main parts: the SS algorithm, SSM establishment, and merging operations. Specifically, for an input spatial tensor, it is first flattened into 1-D sequences along four directions in the H–W plane: from top left to bottom right, bottom right to top left, top right to bottom left, and bottom left to top right. These sequences are then used to establish relational dependencies through (4) and (5). Finally, the processed sequences are merged to produce the output of P-SS2D.

3) Channel Mamba Block: In remote sensing image segmentation tasks, dynamic channel weight allocation is particularly crucial, especially when dealing with spectral feature overlap and color similarity. Objects in different spectral bands often exhibit similar spectral characteristics, leading to color confusion and affecting segmentation accuracy. Mainstream methods, such as SENet [17], CBAM, and UBRFCNet [18] [as shown in Fig. 3(a)], generating weights through global statistical features improves the feature representation ability, but it relies on global spatial statistics, neglects a large amount of local spatial information, and lacks interaction information between different features. This makes it difficult to handle spectral feature overlap, such as the similar reflectance of trees and low vegetation in the near-infrared band, leading to poor segmentation performance. Another category of methods, such as DANet [49], C^{2}Tnet [22], and TCA [50] [see Fig. 3(b)], generate attention matrices by calculating channel dependencies. Although they enhance spatial information, they suffer from high computational complexity and slow speed.

In contrast, our CMB method dynamically models channel information using channel stream modeling, leveraging spatial information. Specifically, we compress spatial and channel features into a single patch and adjust channel weights in real time based on the state of the previous patch. This design localizes the processing of channel–spatial information, allowing the channel weights within each patch to accurately capture local spatial features, thus avoiding the loss of spatial information caused by relying on global statistics in traditional methods. In addition, by using stream modeling at the channel level, CMB effectively enhances interactions between channels, enabling adaptive weight adjustment. This leads to more precise differentiation between land cover types with similar spectral features, significantly improving the accuracy of remote sensing image segmentation.

The module structure of CMB is similar to SMB, but it replaces the P-SS2D algorithm with the C-SS2D algorithm at the channel level, as shown in Fig. 4(b). Specifically, C-SS2D consists of spatial scanning algorithm (CS), SSM establishment, and merging operations. In the CS section, we first apply a 2-D convolution operation with a kernel size equal to the patch size. This operation patches the reshaped features from the preceding fusion patching process along the C–W and C–H dimensions, generating two channel-level tensors as inputs. Subsequently, horizontal forward and reverse, as well as vertical forward and reverse scanning, are performed to

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produce eight sequences. These eight sequences are then used to establish dependencies in the C–W and C–H directions using (4) and (5). Finally, the processed eight output sequences are merged to form the output of CMB. It is worth noting that to achieve dynamic weight updates based on the previous hidden state, we need to weight the first term in (5), as the second term is usually ignored as a skip connection. Weighting the first term in (5) is equivalent to weighting each term in (4). In (4), the second term $\bar{B}$ is solely related to the current input and independent of the previous hidden state, thereby preventing dynamic weight updates based on the prior state. Thus, (4) is reformulated as

$$h_t = \bar{A}(h)_{t-1} + w_t \bar{B} x_t \tag{9}$$

$$w_t = \mathrm{ReLU}(a h_{t-1} + b). \tag{10}$$

Here, w_t is obtained through a linear layer with the previous hidden state h_{t-1}, where a and b are the weights and bias of the linear layer, respectively, and ReLU represents the activation using the ReLU function. Fig. 3(c) illustrates the scanning and dependency establishment in the C–H plane in the horizontal direction, where h_t represents the hidden state at step t in the SSM. Taking the *channel–H* direction as an example, we treat a 2×2 C–H block as a patch input and initialize a weight. The current patch's dependency on the previous patch is established through (5), (9), and (10). Since the value after establishing dependencies for the current patch can be directly inferred from the current patch value and the previous patch's hidden state, there is no need to compute the relationship between each patch and all other patches. The recursive relationship can be efficiently computed in parallel through the associated scanning algorithm, reducing the complexity to linear [55], [59], [60]. Fig. 5 provides a schematic comparison of the time complexity between channel self-attention and C-SS2D. This method decomposes the content of the same channel into multiple patches, increasing the number of weights and achieving a more granular representation. A more granular representation typically captures subtle features and complex relationships more accurately, thus resulting in stronger representation capability through CMB features.

C. Loss Function

The loss function $\mathcal{L}$ used in this article is a combination of a Dice loss $\mathcal{L}_{\text{dice}}$ and a cross-entropy loss function $\mathcal{L}_{\text{ce}}$, which can be expressed as

$$\mathcal{L}_{\text{ce}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} y_k^{(n)} \log \hat{y}_k^{(n)} \tag{11}$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} \frac{\hat{y}_k^{(n)}(y)_k^{(n)}}{\hat{y}_k^{(n)} + y_k^{(n)}} \tag{12}$$

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{dice}} \tag{13}$$

where N represents the number of samples and K represents the number of classes. $y^{(n)}$ and $\hat{y}^{(n)}$ denote the one-hot encoding of the true semantic labels and their corresponding softmax outputs from the network, where $n \in [1, \ldots, N]$. Specifically, $\hat{y}_k^{(n)}$ represents the confidence of sample n belonging to class k.

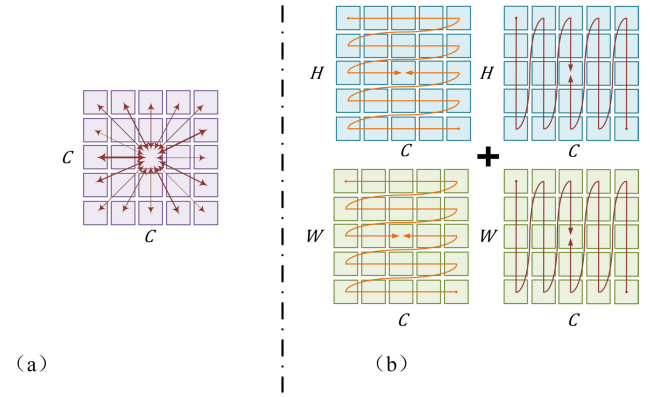


Fig. 5. Comparison of the time complexities of two channel-weighting algorithms. (a) Self-attention-based method for constructing channel attention, which has quadratic complexity. (b) C-SS2D method, which performs scanning along different directions of the C–H and C–W planes and then uses the improved SSM for weighting, exhibiting linear complexity.

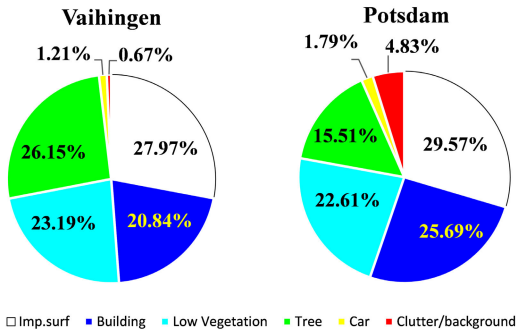


Fig. 6. Datasets' label proportion chart.

IV. Experiment

A. Experimental Settings

1) Datasets: In this study, the experimental phase utilized the Vaihingen and Potsdam aerial datasets, along with the LoveDA satellite dataset, as evaluation benchmarks. These datasets are widely recognized as standard samples in the field of remote sensing image semantic segmentation and are commonly used to assess the performance of remote sensing algorithms, ensuring the general applicability and comparability of the research findings. The three datasets encompass a broad range of land cover categories and complex environmental conditions, including buildings, road systems, and various land cover types, along with seasonal variations, meteorological diversity, and different lighting scenarios. This makes them an ideal testing platform to evaluate the robustness and generalization capabilities of the models. It is worth noting that these datasets have been extensively studied and are widely used in academia, facilitating easy comparison with previous research. Sections IV-A1.a–IV-A1.c provides a detailed description of these three datasets.

a) Vaihingen dataset: This dataset originates from remote sensing images in the Vaihingen area of Germany, consisting of 33 high-resolution TOP image blocks. Each image block has an average size of 2494 × 2064 pixels and contains real orthophoto (TOP), digital surface model (DSM),

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and normalized DSM (NDSM) data. The dataset covers five foreground land cover classes (impervious surfaces, buildings, low vegetation, trees, and cars) and one background land cover class (clutter). In the experiments, we strictly selected training data according to the specific training IDs (IDs 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, and 37) provided by the ISPRS competition, with the remaining 17 images used for testing. This selection ensures consistency with the data of other researchers, facilitating comparison and analysis. For ease of processing, we cropped the image blocks into smaller blocks of size 1024 × 1024 pixels.

b) Potsdam dataset: This dataset uses aerial images from the Potsdam area of Germany, including 38 high-resolution image blocks. Each image block has a ground sampling distance of 5 cm and a size of 6000 × 6000 pixels. Similar to the Vaihingen dataset, each image block consists of real-world object information and a DSM and provides four multispectral bands (red, green, blue, and near infrared). During training, we used specific training IDs provided by the ISPRS competition, including the images numbered 2_10, 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_8, 7_9, 7_11, and 7_12. The remaining 15 images were reserved for testing. We cropped the image blocks into 1024 × 1024 pixel small blocks for analysis. As shown in Fig. 6, the "clutter/background" label constitutes a very small proportion of the Vaihingen and Potsdam datasets. Including this category could lead to an imbalanced impact on the evaluation results, thereby failing to accurately reflect the model's performance on other important categories. Therefore, to ensure the accuracy and fairness of the evaluation results, we opted to ignore the "clutter/background" category.

c) LoveDA dataset: This dataset is constructed using images obtained from the Google Earth platform in three cities in China: Nanjing, Changzhou, and Wuhan. It contains 5987 high-resolution optical remote sensing images (with a ground resolution of 0.3 m), each with a size of 1024 × 1024 pixels, covering seven categories of land cover, including buildings, roads, water, barren land, forests, agriculture, and background [61]. The dataset includes three bands: red, green, and blue. Unlike the processing of the Vaihingen and Potsdam datasets, the LoveDA dataset's training, validation, and test sets are divided based on geographical locations to increase differences between regions. Specifically, 2522 images are used for training, 1669 images are used for validation, and 1796 images are provided by the official source for testing. The dataset covers two scenarios: urban and rural. The presence of multiscale objects, complex backgrounds, and inconsistent class distributions in the dataset introduces considerable challenges.

2) Implementation Details: In our study, we utilized the Ubuntu 18.04 operating system and deployed all models on a single NVIDIA GeForce RTX 2080 Ti 11 GB GPU using the PyTorch 1.11 framework. To expedite the convergence speed of the models, we employed the AdamW optimizer with a base learning rate set to $6e^{-4}$ and utilized a cosine learning rate scheduling strategy. For the Vaihingen and Potsdam datasets, we conducted the following data preprocessing steps. First, we randomly cropped images into patches of size 1024 × 1024. During the training phase, we employed various data augmentation techniques, including random scaling (scales of 0.5, 0.75, 1.0, 1.25, and 1.5), random vertical flipping, random horizontal flipping, and random rotation. The training process involved a total of 105 epochs. During the testing phase, we employed multiscale evaluation and random flipping augmentation techniques to ensure the robustness and performance of the models.

3) Evaluation Metrics: This study adopts commonly used performance evaluation metrics in the field of remote sensing segmentation, including overall accuracy (OA), $F1$ Score, and mean intersection over union (mIoU). In addition, to assess the model's parameter quantity, we introduce the parameters and FLOPs metrics. Before explaining these metrics, it is necessary to understand some related terms and symbols: tp (true positives), fp (false positives), fn (false negatives), and tn (true negatives).

a) Precision: This metric measures the proportion of true positives among all samples predicted as positives by the model. In simple terms, precision tells us the accuracy of the model when judging a sample as positive

$$\text{Precision} = \frac{\text{tp}}{\text{tp} + \text{fp}}. \tag{14}$$

b) Recall: It refers to the proportion of samples predicted as positives by the model among all true positive samples. It measures the model's ability to discover all positives

$$\text{Recall} = \frac{\text{tp}}{\text{tp} + \text{fn}}. \tag{15}$$

c) Overall Accuracy: It is one of the commonly used performance evaluation metrics in image classification tasks, indicating the proportion of correctly classified samples to the total number of samples

$$\text{OA} = \frac{\text{tp} + \text{tn}}{\text{tp} + \text{fp} + \text{fn} + \text{tn}}. \tag{16}$$

d) F1 Score: The $F1$ Score is the harmonic mean of precision and recall, considering both the model's accuracy and its ability to capture positives. For multiclass problems, $F1$ Score is typically calculated for each class and then averaged

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}. \tag{17}$$

Overall $F1$ Score = Average of $F1$ Scores for all classes.

e) Mean Intersection Over Union: It is a commonly used evaluation metric in semantic segmentation tasks, measuring the accuracy of the model in pixel-level segmentation. It calculates the average IoU for all classes, where IoU measures the model's segmentation results for each class

$$\text{IoU} = \frac{\text{tp}}{\text{tp} + \text{fp} + \text{fn}}. \tag{18}$$

mIoU is the sum of IoU values for all categories divided by the number of categories.

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B. Ablation Experiment

We conducted a series of experiments to evaluate the performance of different components of CPSSNet. These experiments were validated using the Vaihingen, Potsdam, and LoveDA datasets. In our discussion, we primarily focus on two performance metrics: mIoU and meanF1. All experimental results are averaged over multiple trials.

1) Components of CPSSNet: Table I presents the results after removing individual modules, where CMB refers to the channel visual SSM and SMB refers to the position visual SSM. In the experiments, removed components are denoted as (w/o), while added components are marked as (+). Fig. 7 illustrates the segmentation performance following the removal of individual modules from CPSSNet. As shown in Fig. 8, the removal of any component from CPSSNet adversely impacts the network's performance. To further assess the effectiveness of CMB and SMB, we conducted experiments by adding each module individually to the Vaihingen dataset. The baseline model consists of two convolutional layers added directly after the feature extraction network, ConvNeXt, followed by the segmentation head to obtain segmentation results. For each additional component, the corresponding module is inserted directly before the segmentation head. Figs. 8 and 9 show the segmentation results after sequentially adding each component to the baseline. Overall, the results demonstrate that each proposed component significantly enhances the model performance.

Table III presents the experimental results of replacing different backbones. Furthermore, to assess the robustness of the model, we evaluated the impact of different patch sizes and various parameter settings for the CMB and SMB modules on segmentation performance. The experimental results are presented in Table IV. Different parameter configurations are denoted as "parameter standard." The results indicate that the effects of varying parameter standards and patch sizes on segmentation performance are minimal. The overall trend suggests that as the patch size decreases, the number of patches increases, while the parameter count per patch decreases, resulting in higher computational complexity but improved accuracy. Conversely, when the number of channels per patch increases, both the parameter count and computational complexity rise, but so does the accuracy. Notably, when the patch size is set to 8×8 and parameter standard 1 is applied, CPSSNet achieves optimal segmentation performance, with both the parameter count and computational complexity remaining within acceptable limits. As a result, this configuration is chosen as the default setting for CPSSNet, ensuring optimal performance while effectively balancing computational resource usage.

2) Impact of CMB: As shown in Table I, after removing CMB from CPSSNet on the Vaihingen dataset, the mIoU and $F1$ decreased by 0.95% and 0.54%, respectively, while on the Potsdam dataset, they decreased by 0.93% and 0.6%, respectively. On the other hand, adding CMB to the baseline modeled to an increase of 0.95% in mIoU and 0.69% in $F1$ on the Vaihingen dataset, and an increase of 0.86% in mIoU and 0.64% in $F1$ on the Potsdam dataset. These results clearly demonstrate that CMB has a positive impact on model performance, contributing to at least a 0.86% improvement in mIoU and a 0.54% improvement in $F1$.

Through the visualization results in Fig. 8, we observe that after introducing CMB into the baseline model, it shows significant advantages in feature representation for land cover types with high color similarity (such as trees and low vegetation

TABLE I
RESULTS OF CPSSNET WITH INDIVIDUAL COMPONENTS REMOVED

Method	Params(K)↓	Flops(M)↓	mIoU(%)↑	F1(%)↑
SENet	32.77	4.01	84.11	91.02
BAM-CAM	8.50	**0.83**	84.25	91.19
DANet-CAM	197.38	14.86	84.30	91.26
AFCA	65.797	0.84736	84.34	91.22
CASB	612.971	732.21	84.45	91.29
CMAN-CAM	3.41	4.25984	84.20	91.16
C-SS2D(Ours)	**0.32**	8.92	**84.72**	**91.63**

The best results are indicated in bold black.In the table header, an upward arrow indicates that a higher evaluation value is better, and a downward arrow indicates that a higher evaluation value is better.

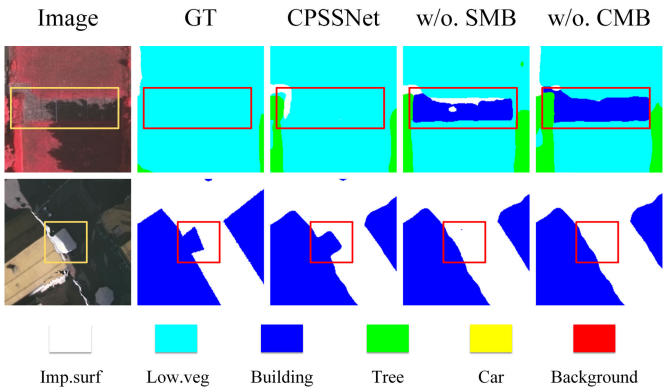


Fig. 7. Local enlarged views of segmentation results from various components removed in CPSSNet. It can be observed from the figures that removing any component leads to a decrease in segmentation performance.

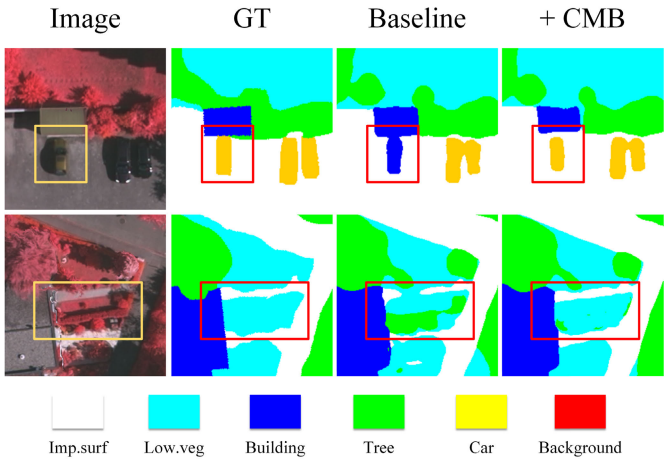


Fig. 8. Locally enlarged view of the segmentation results after incorporating the CMB structure into the baseline model. From the figure, it can be observed that the addition of the CMB structure significantly improves the model's segmentation performance when handling objects with similar colors.

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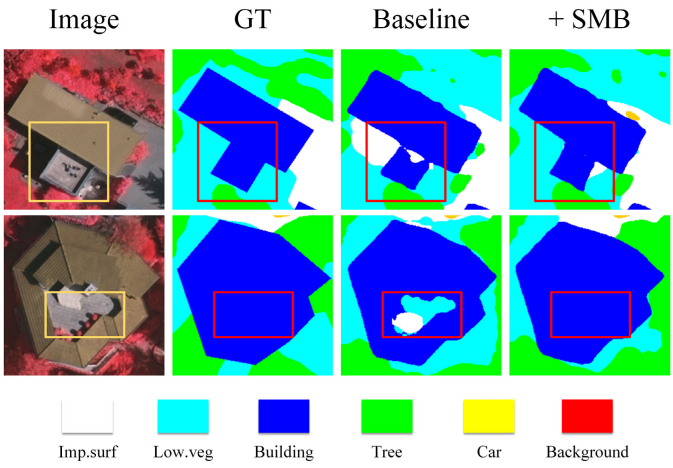


Fig. 9. Zoomed-in results of segmentation with the addition of the SMB structure in the baseline model. It can be observed from the figure that the model performs better in segmenting large continuous objects after the addition of SMB.

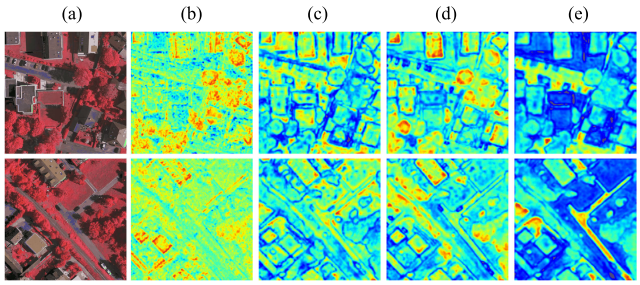


Fig. 10. Feature visualization of two channel modeling methods is provided. (a) Remote sensing image. (b) Features before channel modeling. (c) Features after channel modeling using the BAM method. (d) Features after channel modeling using the DANet method. (e) Features weighted by the SSM using C-SS2D.

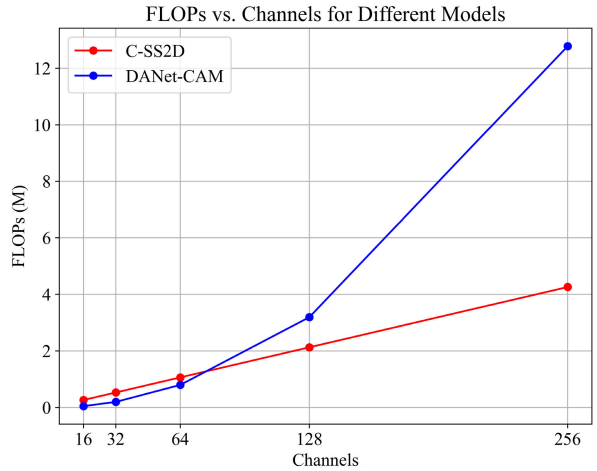


Fig. 11. FLOPs of the self-attention-based channel attention method and the C-SS2D method as the number of channels varies. We tested the FLOPs for channel numbers of 16, 32, 64, 128, and 256.

classes) compared to traditional channel-weighting methods. The dynamic weight adjustment mechanism of CMB effectively distinguishes subtle differences between similar features, enhancing the model's adaptability to complex scenes. This method not only improves land cover identification but also enhances segmentation accuracy, fully demonstrating its professionalism and effectiveness when dealing with highly similar land cover types.

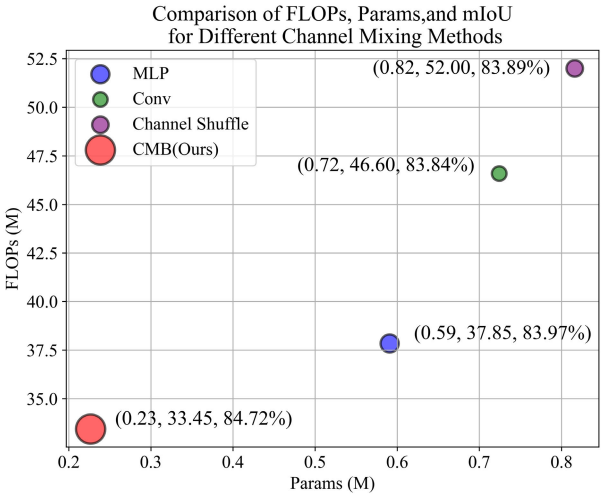


Fig. 12. Parameters, FLOPs, and mIoU of several mainstream channel mixing methods.

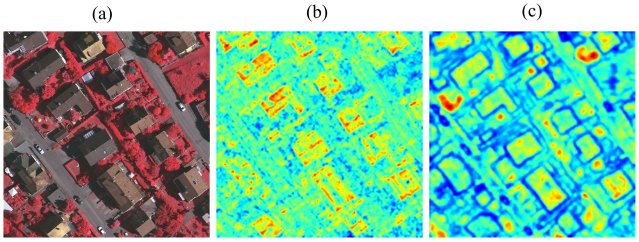


Fig. 13. Feature maps before and after using CMB. (a) Remote sensing image. (b) Feature map without using CMB. (c) Feature map after applying CMB. It can be clearly observed that after using CMB, the boundaries between different land cover categories in the feature map become more distinct, and the discriminative ability is significantly improved.

TABLE II
COMPARISON OF DIFFERENT CHANNEL MODELING APPROACHES

Method	Vaihingen		Potsdam	
	mIoU(%)	F1(%)	mIoU(%)	F1(%)
CPSSNet	84.72	91.63	87.46	93.21
CPSSNet w/o CMB	83.77	91.09	86.53	92.61
CPSSNet w/o SMB	83.82	91.11	86.55	92.66
CPSSNet w/o $\mathcal{L}_{dice}$	84.53	91.53	87.12	93.13
CPSSNet w/o $\mathcal{L}_{ce}$	84.24	91.36	86.92	92.78
Baseline	82.62	90.59	85.61	91.53
Baseline + CMB	83.57	91.28	86.47	92.17
Baseline + SMB	83.65	91.33	86.52	92.21

To more intuitively demonstrate the impact of the CMB on features, we provide visualizations of the feature maps before and after using CMB, as shown in Fig. 13. Fig. 13(a) displays the feature map without CMB, while Fig. 13(b) shows the feature map after applying CMB. The comparison clearly shows that after using CMB, the differentiation between different land cover types in the feature map is

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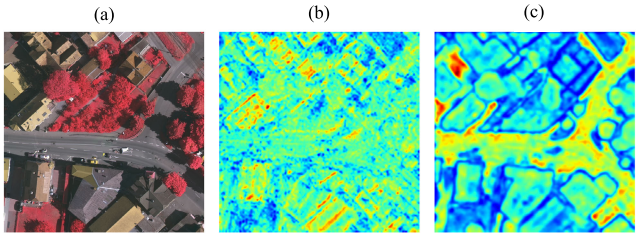


Fig. 14. Feature maps before and after using SMB. (a) Remote sensing image. (b) Features before using SMB. (c) Features after using SMB. It can be observed that the large, continuous objects in the feature map after using SMB exhibit higher response values.

TABLE III
COMPARISON OF DIFFERENT BACKBONES

Method	Backbone	Params(M)↓	mIoU(%) ↑
CPSSNet	ResNet50	25.56	84.31
	ResNext50	25.03	84.44
	ResNest50	25.48	84.53
	Maxvit-Tiny	31.05	84.67
	Swinv2-Tiny	28.33	84.52
	Vim-Small	25.87	84.61
	Vmamba-Tiny	22.37	84.52
	ConvNext-Tiny	28.59	84.72

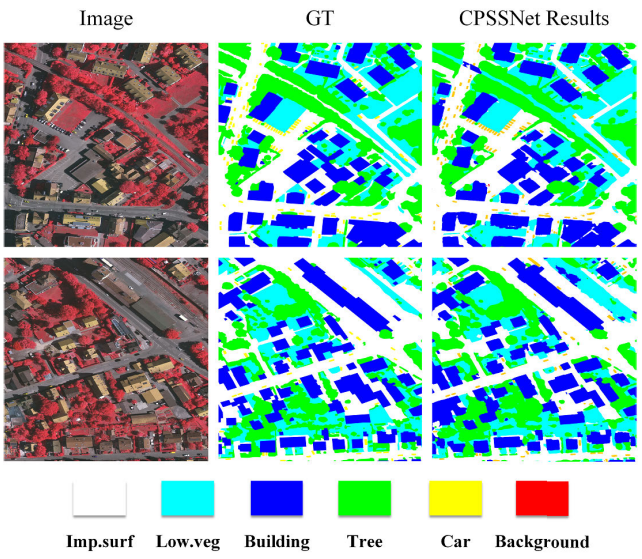


Fig. 15. Segmentation results of IDs 8 and 10 on the Vaihingen dataset. The first column displays the original images, the second column shows the ground truth (GT), and the third column presents the segmentation results of CPSSNet.

significantly improved, with boundaries becoming clearer. Specifically, CMB enhances the feature differences between similar land cover classes by applying weights to different channels, thereby improving the model's ability to handle complex scenes. In particular, for land cover types with color similarities, CMB effectively enhances feature representation, leading to a significant improvement in segmentation accuracy.

To demonstrate the advantages of the core algorithm C-SS2D in CMB over other channel direction modeling methods, we replaced the C-SS2D in the CMB block of CPSSNet with several other channel modeling methods and conducted related experiments and metric tests. Since this section only considers channel modeling, we have omitted the spatial modeling component of simultaneous channel–spatial modeling. We have appended the module name (-module name) to indicate this omission. We compared several channel modeling methods, including the SEM from SENet [17], the channel attention module (CAM) from BAM [21], the CAM from DANet [49], the AFGCA from UBRFCNet [18], the channel-aware SwinT block (CASB) from C^{2}TNet [22], and the channel–spatial attention block (CSAB) from CMAN [20] The experimental results are shown in Table II. As the results indicate, C-SS2D's FLOPs are not low due to the need to compute hidden states, but its parameters are the lowest. This is because C-SS2D does not need to maintain a weight tensor for the channel direction; instead, the weight tensors in C-SS2D are computed dynamically, and C-SS2D only retains the mapping weights used for the computations. In addition, we compared several mainstream channel mixing methods, including MLP, Conv, and Channel Shuffle. The results are shown in Fig. 12. As can be seen from the figure, CMB demonstrates the best performance in terms of mIoU and parameter count. However, in terms of FLOPs, there is still a noticeable gap compared to the two Conv methods.

Fig. 10 provides the visualizations of features obtained with two different channel modeling methods. In Fig. 10(a), the features before channel modeling are shown, while Fig. 10(b) and (c) show features after channel modeling using BAM and DANet, respectively. Fig. 10(d) presents features weighted by C-SS2D utilizing the SSM. It is evident that features weighted by C-SS2D exhibit clearer structural contours and richer color contrasts, providing better information for segmentation tasks. From the metrics, it is evident that C-SS2D performs better, indicating that features processed by C-SS2D have better representation capability.

Furthermore, to verify that CMB has linear complexity compared to DANet's channel self-attention method, we varied the channel dimension of the input tensor and tested its FLOPs. The input tensor size was (32, 32). The experimental results in Fig. 11 show that DA-C clearly exhibits quadratic complexity, while C-SS2D demonstrates linear complexity. This experiment confirms that C-SS2D has linear complexity.

3) Impact of SMB: As shown in Table I, for CPSSNet, removing the SMB results in a decrease of 0.9% in mIoU and 0.52% in $F1$ on the Vaihingen dataset, and a decrease of 0.91% in mIoU and 0.55% in $F1$ on the Potsdam dataset. On the other hand, adding the SMB to the baseline improves the mIoU by 1.03% and $F1$ by 0.73% on the Vaihingen dataset and improves the mIoU by 0.91% and $F1$ by 0.68% on the Potsdam dataset. These results demonstrate the positive impact of SMB on the segmentation model, effectively enhancing the segmentation accuracy.

Fig. 9 displays the segmentation results obtained by adding SMB to the baseline. It can be seen from the results that adding SMB to the baseline significantly enhances the segmentation of large continuous objects (e.g., buildings), which is typically a characteristic of global features. To further verify the

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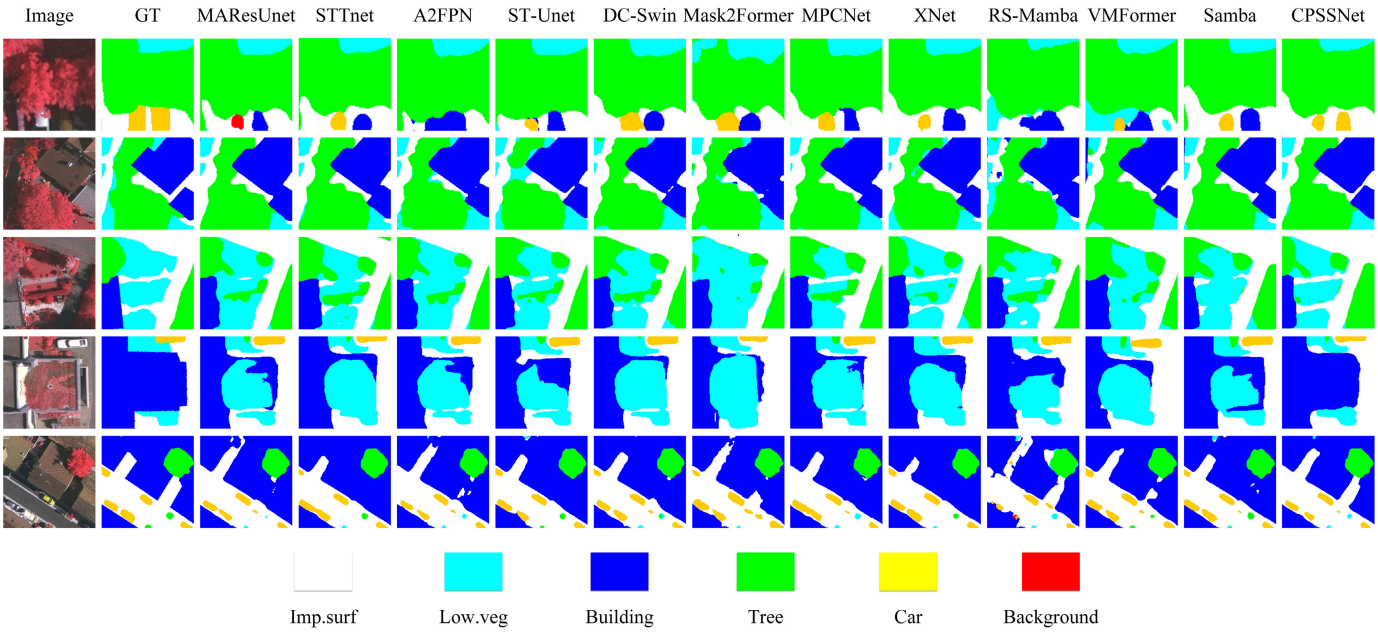


Fig. 16. Local zoomed-in view of segmentation results on the Vaihingen dataset for CPSSNet and other comparison models. From the figure, it can be observed that our model achieves better results in segmenting textures, large continuous objects, and elongated continuous objects.

TABLE IV
IMPACT OF DIFFERENT PARAMETER CONFIGURATIONS

Patch Size	SSM Parameters							
	Parameter Standard 1				Parameter Standard 2			
	CMB		SMB		CMB		SMB	
2×2	[linear : $2\times96\rightarrow4\times96$ DWConv : $3\times3, 4\times96$ C-SS2D dim : 4×96 linear : $4\times96\rightarrow2\times96$]		[linear : $96\rightarrow2\times96$ DWConv : $3\times3, 2\times96$ C-SS2D dim : 2×96 linear : $2\times96\rightarrow96$]		[linear : $2\times128\rightarrow4\times128$ DWConv : $3\times3, 4\times128$ C-SS2D dim : 4×128 linear : $4\times96\rightarrow2\times128$]		[linear : $128\rightarrow2\times128$ DWConv : $3\times3, 2\times128$ C-SS2D dim : 2×128 linear : $2\times128\rightarrow128$]	
	Params(M)	FLOPs(G)	mIoU(%)		Params(M)	FLOPs(G)	mIoU(%)	
	28.97	125.60	84.70		29.83	171.72	84.68	
8×8	[linear : $2\times96\rightarrow4\times96$ DWConv : $3\times3, 4\times96$ C-SS2D dim : 4×96 linear : $4\times96\rightarrow2\times96$]		[linear : $96\rightarrow2\times96$ DWConv : $3\times3, 2\times96$ C-SS2D dim : 2×96 linear : $2\times96\rightarrow96$]		[linear : $2\times128\rightarrow4\times128$ DWConv : $3\times3, 4\times128$ C-SS2D dim : 4×128 linear : $4\times96\rightarrow2\times128$]		[linear : $128\rightarrow2\times128$ DWConv : $3\times3, 2\times128$ C-SS2D dim : 2×128 linear : $2\times128\rightarrow128$]	
	Params(M)	FLOPs(G)	mIoU(%)		Params(M)	FLOPs(G)	mIoU(%)	
	31.73	110.65	84.72		34.75	126.72	84.63	
32×32	[linear : $2\times96\rightarrow4\times96$ DWConv : $3\times3, 4\times96$ C-SS2D dim : 4×96 linear : $4\times96\rightarrow2\times96$]		[linear : $96\rightarrow2\times96$ DWConv : $3\times3, 2\times96$ C-SS2D dim : 2×96 linear : $2\times96\rightarrow96$]		[linear : $2\times128\rightarrow4\times128$ DWConv : $3\times3, 4\times128$ C-SS2D dim : 4×128 linear : $4\times96\rightarrow2\times128$]		[linear : $128\rightarrow2\times128$ DWConv : $3\times3, 2\times128$ C-SS2D dim : 2×128 linear : $2\times128\rightarrow128$]	
	Params(M)	FLOPs(G)	mIoU(%)		Params(M)	FLOPs(G)	mIoU(%)	
	75.98	110.36	84.47		113.39	123.91	84.54	

global modeling capability of SMB, we directly compared the features before and after using SMB in CPSSNet, as shown in Fig. 14, where Fig. 14(a) represents the features before using SMB and Fig. 14(b) represents the features after using SMB. It is evident from the figure that after using SMB, large continuous regions have higher values, while small discontinuous regions have lower values, indicating that SMB indeed establishes remote dependencies effectively, thus confirming its effective global modeling capability.

4) Impact of Backbone: To eliminate the impact of the Backbone on the results, we conducted multiple backbone replacement experiments on the Vaihingen dataset, selecting several widely used backbones with similar parameter scales. These included convolutional networks (such as ConvNext-Tiny, ResNet50, ResNext50, and ResNest50), Transformer-based networks such as Maxvit-Tiny and Swinv2-Tiny, as well as Mamba-based structures such as Vim-Small [35] and Vmamba-Tiny [36]. The experimental results are shown in Table III, where all these backbones demonstrated high-performance metrics. In terms of parameter count, Vmamba-Tiny had the smallest number of parameters across all experiments, showing better computational efficiency. Nevertheless, considering both parameter count and accuracy, ConvNext-Tiny provided the best performance while maintaining a low parameter count and is thus recommended as the preferred Backbone. Although Vim-Small and Vmamba-Tiny have advantages in terms of parameter count, their accuracy is slightly inferior to that of ConvNext-Tiny.

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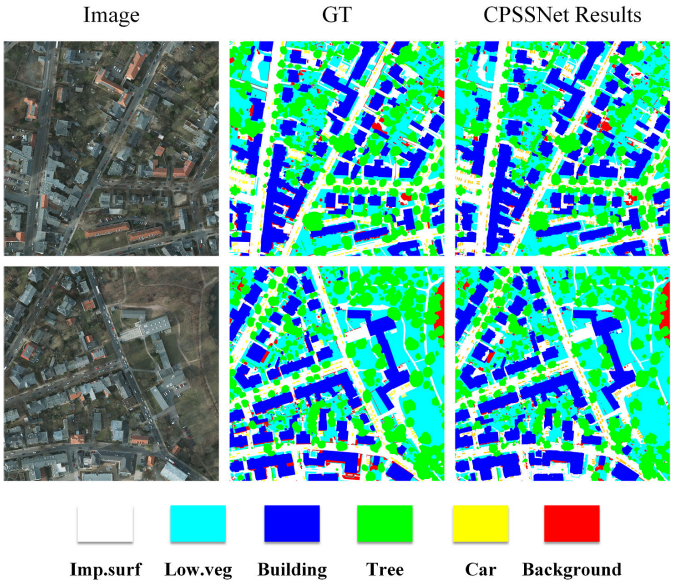


Fig. 17. Segmentation results of ID 3_13 and 3_14 on the Potsdam dataset. The first column represents the original image, the second column shows the ground truth (GT), and the third column displays the segmentation results obtained using CPSSNet.

5) Impact of the Loss Function: From the experimental results in Table I, it can be observed that different configurations of the loss function have varying degrees of impact on model performance. On the Vaihingen dataset, removing the Dice loss resulted in a decrease of 0.19% in mIoU and 0.10% in $F1$, while removing the cross-entropy loss led to a reduction of 0.48% in mIoU and 0.27% in $F1$. Similarly, on the Potsdam dataset, removing $\mathcal{L}_{\text{dice}}$ had a smaller effect, with mIoU decreasing by 0.34% and $F1$ by 0.08%, whereas removing $\mathcal{L}_{\text{ce}}$ had a more significant impact, with mIoU decreasing by 0.54% and $F1$ by 0.43%. These results indicate that $\mathcal{L}_{\text{dice}}$ has a relatively mild effect on the model, primarily affecting class imbalance and segmentation of small objects. The overall performance drop after its removal is relatively small. On the other hand, L_{ce} has a more pronounced impact on the model, especially on the Potsdam dataset, where removing cross-entropy loss significantly affects the model's classification ability, with notable reductions in both mIoU and $F1$. Overall, the combination of $\mathcal{L}_{\text{dice}}$ and $\mathcal{L}_{\text{ce}}$ achieves a better balance between segmentation accuracy and class discrimination ability. Removing either of these loss functions has a certain negative effect on model performance, especially in scenes with complex classes and rich details.

C. Comparative Experiments

For the comparison models, we selected the following: the multistage attention residual UNet (MAResU-Net) with linear attention mechanism [41], the attention aggregated feature pyramid convolutional network (A2-FPN) [63], the Swin-Transformer network with a dense connection feature aggregation module (DCFAM), known as DC-Swin [65], the Transformer network Mask2Former that performs attention using masks [66], the dual-branch segmentation network combining Transformer and CNN (ST-Unet) [64], the network with a multiscale prototype-based Transformer decoder (MPCNet) [67], sparse token transformers (STTs) with sparse label sampler (STTNet) [62], the XNet [68] network that directly utilizes frequency-domain features for segmentation, the RS-Mamba [38] and Samba [40], which also uses SSM for dense prediction tasks, and the Transformer network VMFormer with variable window attention [69]. Our model ultimately achieved higher accuracy on the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets, which are widely used for remote sensing segmentation tasks, compared to the aforementioned models. Fig. 20 shows the shortcomings of CPSSNet on the LoveDA dataset. Tables VI–VIII represent the comparison results of CPSSNet and other models on the Vaihingen dataset, Potsdam dataset, and LoveDA dataset, respectively.

1) Results of the Vaihingen Dataset: Table V presents the experimental results of various comparison methods and CPSSNet on the Vaihingen dataset. The results indicate that our proposed CPSSNet achieves significant improvement over other networks in terms of average $F1$, mIoU, and OA, reaching 91.63%, 84.72%, and 91.81%, respectively (Table VI). Specifically, CPSSNet outperforms excellent networks such as the pure convolutional networks MAResU-Net and A2-FPN, as well as attention-based network MPCNet. In addition, CPSSNet surpasses the Transformer network DC-Swin, the CNN-Transformer hybrid network ST-Unet, and the frequency-domain segmentation method XNet. Moreover, compared to RS-Mamba, which also utilizes SSM, our CPSS-Net exceeds RS-Mamba by 1.04% in $F1$, 1.76% in mIoU, and 1.06% in OA. The experimental results demonstrate that our network exhibits superior segmentation performance.

In addition to comparing the segmentation accuracy of each model, we also compared the convergence speed of the proposed models and CPSSNet. The results are shown in Fig. 21, which illustrates the trend of mIoU variation with epoch during training on the Vaihingen dataset. We selected the first 44 rounds for reference, with the 15th epoch marking the end of the first cycle of cosine annealing strategy. As shown in Fig. 21, our model converges fastest, indicating that CPSSNet has superior convergence performance compared to other networks. Furthermore, compared to other models, our model exhibits smaller fluctuations, suggesting that our model can more stably learn the features of tasks and datasets during training.

Fig. 15 presents the segmentation results for IDs 8 and 10 from the Vaihingen dataset. Compared to other Transformer and CNN-based networks (see Fig. 16), CPSSNet demonstrates significant advantages in segmenting large, continuous objects (such as buildings) and land cover types with color similarity. This highlights the model's robust global modeling and dynamic channel modeling capabilities. The dynamic weight allocation mechanism for stream channel information plays a crucial role in handling objects with high color similarity while also enhancing the model's ability to capture fine structures such as roads. This allows CPSSNet to accurately distinguish between objects with similar colors, thereby improving segmentation accuracy. Overall, CPSSNet outperforms other networks in its ability to capture intricate details in complex scenes, showcasing its strength and effectiveness in handling objects with high color similarity.

In addition, to comprehensively evaluate the model's time performance, we tested the parameter count (Params), FLOPs,

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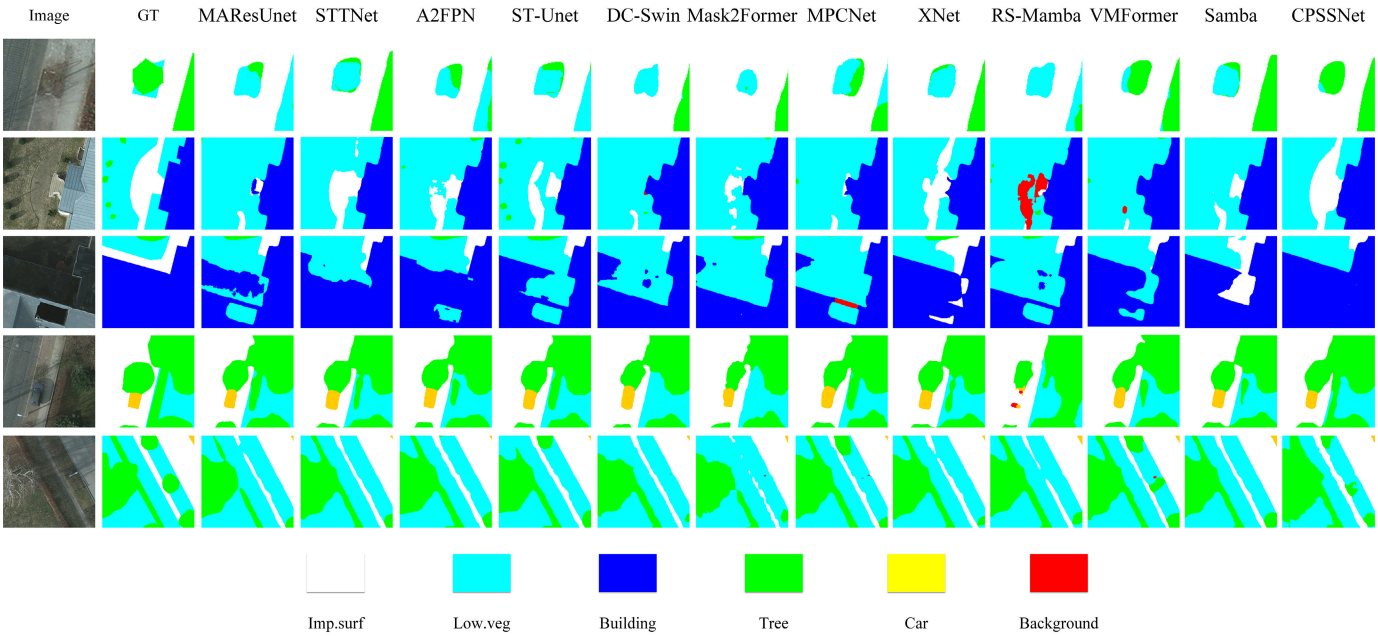


Fig. 18. Local zoomed-in view of segmentation results on the Potsdam dataset for CPSSNet and other comparative models. Our model exhibits better results for segmenting textures, large continuous objects, and elongated continuous objects compared to other models on the Vaihingen dataset. In addition, CPSSNet demonstrates better performance on categories with certain consistency across classes on this dataset.

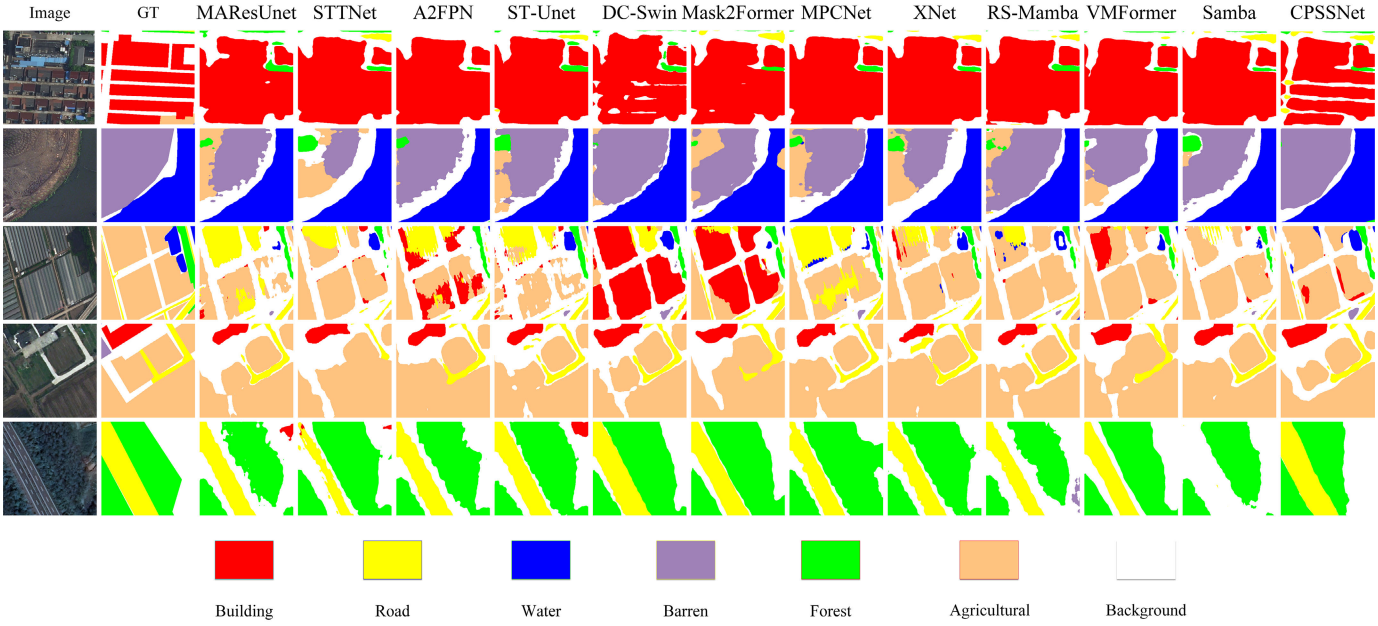


Fig. 19. Local zoomed-in view of segmentation results on the LoveDA dataset for CPSSNet and other comparative models. Our model exhibits better results for segmenting textures, large continuous objects, and elongated continuous objects compared to other models on the Vaihingen dataset. In addition, CPSSNet demonstrates better performance on categories with certain consistency across classes on this dataset.

and inference time for a single image of CPSSNet. The experimental results are shown in Table VIII. According to the data in Table VIII, for images of size 1024 × 1024, CPSSNet demonstrates a significant advantage in FLOPs compared to lightweight convolutional neural networks such as MAResUnet, A2-FPN, and MPCNet, and is also superior to Transformer-based networks such as STUnet, DC-Swin, mask2former, and VMFormer. This advantage is attributed to the linear complexity of the SSM, which allows CPSSNet to maintain low FLOPs when processing 1024 × 1024 size images. CPSSNet's inference time is 147.54 ms, showing superior efficiency, particularly when compared to models with higher computational complexity such as Xnet and MPCNet. This makes CPSSNet capable of maintaining low computational overhead when processing high-resolution remote sensing images, thus enhancing the model's practical applicability. Although the inference time is slightly longer than that of some lightweight models (such as MAResUnet), CPSSNet still demonstrates a strong advantage in inference efficiency compared to other models, making it

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TABLE V
COMPARISON OF SEGMENTATION RESULTS ON THE VAIHINGEN DATASET

Method	F1(%)					Evaluation index		
	Imp.Surf	Building	Lowveg	Tree	Car	MeanF1(%)	mIoU(%)	OA(%)
MAResU-Net[41]	92.91	95.26	84.95	89.94	88.33	90.28	83.30	90.86
STTNet[62]	92.84	95.17	83.35	90.61	86.92	89.78	81.71	90.67
A2-FPN[63]	92.99	95.53	84.67	90.34	87.62	90.23	82.42	91.04
ST-Unet[64]	92.00	94.85	83.87	90.88	86.40	89.60	81.39	90.53
DC-Swin[65]	93.46	96.00	85.32	90.03	84.88	89.94	82.01	91.29
Mask2Former[66]	92.86	96.03	84.15	90.50	89.30	90.57	82.99	91.10
MPCNet[67]	92.76	95.50	84.70	90.40	90.44	90.76	83.27	90.93
XNet[68]	92.89	95.48	85.17	91.10	88.22	90.57	82.95	91.21
RS-Mamba[38]	93.10	95.61	84.09	90.26	89.42	90.50	82.87	90.98
VMFormer[69]	93.23	95.81	85.64	91.08	88.65	90.88	83.48	91.54
Samba[40]	93.03	95.44	83.84	90.70	89.11	90.42	82.75	90.99
CPSSNet(Ours)	**93.51**	**96.09**	**85.97**	**91.23**	**91.33**	**91.63**	**84.72**	**91.82**

The best results are indicated in bold black.

TABLE VI
COMPARISON OF SEGMENTATION RESULTS ON THE POTSDAM DATASET

Method	F1(%)					Evaluation index		
	Imp.Surf	Building	Lowveg	Tree	Car	MeanF1(%)	mIoU(%)	OA(%)
MAResU-Net[41]	92.38	95.83	86.65	88.44	96.13	91.89	85.22	90.28
STTNet[62]	92.02	95.99	88.24	89.23	95.14	92.12	85.55	92.12
A2-FPN[63]	92.85	95.84	87.17	88.76	96.13	92.15	85.66	90.68
ST-Unet[64]	92.77	96.90	86.45	88.11	95.80	92.01	85.46	90.53
DC-Swin[65]	93.26	96.86	87.74	88.68	95.50	92.41	86.10	91.16
Mask2Former[66]	92.48	96.41	87.53	89.37	96.08	92.37	86.03	90.83
MPCNet[67]	92.69	96.38	87.30	88.74	96.34	92.29	85.91	90.56
XNet[68]	91.98	96.80	86.93	88.76	95.81	92.05	85.51	90.56
RS-Mamba[38]	92.56	96.63	87.43	88.51	95.72	92.17	85.70	90.70
VMFormer[69]	92.88	96.22	88.47	89.72	95.75	92.61	86.39	91.26
Samba[40]	92.87	96.99	87.49	89.16	95.74	92.45	86.17	91.05
CPSSNet(Ours)	**93.49**	**97.05**	**88.81**	**90.05**	**96.62**	**93.21**	**87.46**	**91.76**

The best results are indicated in bold black.

especially suitable for providing efficient solutions in high computational demand scenarios. While CPSSNet performs excellently in terms of FLOPs and inference time, its overall parameter count is not the most optimized. In the future, we will investigate this aspect and perform further optimization.

2) Results on the Potsdam Dataset: To comprehensively evaluate the segmentation performance of CPSSNet, we further conducted comparative experiments on the Potsdam dataset. The data results in Table VI demonstrate that similar to the Vaihingen dataset, CPSSNet outperforms other networks on the Potsdam dataset. Specifically, our $F1$ Score reached 93.21%, mIoU reached 87.46%, and OA reached 91.76%. Compared to RS-Mamba, which also utilizes SSM, CPSSNet achieved higher $F1$, mIoU, and OA scores by 1.04%, 1.76%, and 1.06%, respectively. Particularly noteworthy is that CPSSNet achieved a higher $F1$ Score for the Lowveg class on the Potsdam dataset, surpassing the highest $F1$

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TABLE VII
COMPARISON OF SEGMENTATION RESULTS ON THE LOVEDA DATASET

Method	F1(%)							Evaluation index		
	Building	Road	Water	Barren	Forest	Agricultural	Background	MeanF1(%)	mIoU(%)	OA(%)
MAResU-Net[41]	77.75	70.86	77.10	38.59	61.05	68.58	68.34	66.04	50.45	68.92
STTNet[62]	73.40	67.77	82.13	45.41	58.65	62.14	66.30	65.11	49.21	66.41
A2-FPN[63]	74.45	67.67	73.11	**56.04**	59.47	69.98	68.49	67.03	50.74	68.73
ST-Unet[64]	76.53	72.15	82.16	46.44	58.76	66.03	67.02	67.01	51.38	68.16
DC-Swin[65]	72.71	72.28	83.30	49.23	57.42	62.17	69.00	66.59	50.83	68.01
Mask2Former[66]	77.51	72.59	83.34	44.07	60.97	68.16	70.93	68.23	52.92	70.82
MPCNet[67]	78.82	71.16	83.45	46.44	55.55	70.38	70.76	68.08	52.80	70.87
XNet[68]	**79.35**	**72.88**	**83.78**	38.35	59.34	70.60	71.27	67.94	53.00	71.40
RS-Mamba[38]	78.79	72.50	83.21	43.61	**61.52**	69.41	70.88	68.56	53.35	71.16
VMFormer[69]	78.50	72.60	83.53	50.18	56.11	70.10	71.40	68.92	53.61	71.47
Samba[40]	76.01	67.84	81.98	46.37	59.43	66.98	69.61	66.89	51.19	69.19
CPSSNet(Ours)	78.49	72.69	82.80	55.02	59.52	**74.25**	**71.75**	**70.65**	**55.38**	**72.83**

The best results are indicated in bold black.

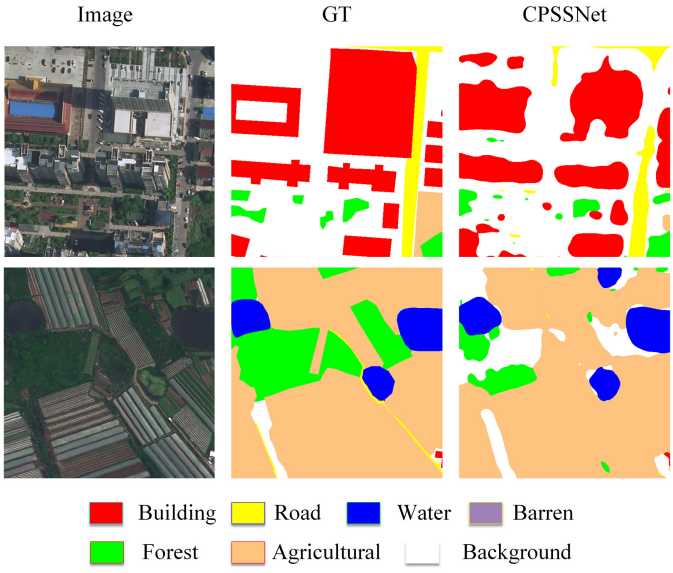


Fig. 20. Segmentation results of CPSSNet on the LoveDA dataset: the first column shows the original images, the second column shows the ground truth (GT), and the third column shows the CPSSNet results.

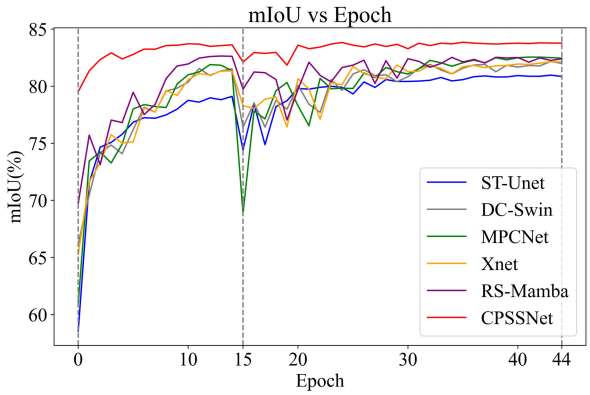


Fig. 21. Change in mIoU over epochs for CPSSNet and several mainstream segmentation methods. It can be observed from the graph that CPSSNet converges faster and exhibits greater stability compared to other networks.

Score achieved by DC-Swin by 1.07%. This is attributed to the inherent consistency within the Lowveg class, which the channel dynamic weighting mechanism of CPSSNet can better leverage. Due to differences in dataset size and type, segmentation accuracy on the Potsdam dataset is generally higher than that on the Vaihingen dataset.

We also provide the segmentation results for IDs 3_13 and 3_14, as shown in Fig. 17. Fig. 18 illustrates the segmentation results of various methods on the Potsdam dataset, as mentioned in Table VI. Similar to the performance on the Vaihingen dataset, CPSSNet demonstrates superior performance in segmenting large, continuous objects, further validating its effective long-range dependency modeling capability. Compared to other Transformer and CNN networks, CPSSNet excels at handling long, continuous objects, highlighting the effectiveness of dynamic channel weight allocation in addressing color similarity. In addition, the first, fourth, and fifth rows of Fig. 18 show that CPSSNet performs well in segmenting objects with high intraclass consistency and small interclass differences. These results emphasize the advantages of our method in complex scenes, particularly in dealing with land cover types that exhibit strong color similarity.

3) Results on the LoveDA Dataset: To further validate the generalizability of CPSSNet, we conducted comparative experiments on the satellite dataset LoveDA. As shown in Table VIII, CPSSNet outperforms other networks in overall performance on the LoveDA dataset. Notably, CPSSNet significantly outperforms all other networks in the agricultural category. This is because the agricultural class in the LoveDA dataset contains the most continuous structures, and CPSSNet

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TABLE VIII
COMPARISON OF PERFORMANCE ACROSS DIFFERENT MODELS

Method	Image Size	Params(K)	FLOPs(M)	Inference Time(ms)
MAResUnet[41]	1024^2	26.28	140.12	**50.14**
STTNet[62]	1024^2	146.22	212.59	799.94
A2-FPN[63]	1024^2	**22.82**	167.33	176.62
STUnet[64]	1024^2	75.48	385.55	344.42
DC-Swin[65]	1024^2	66.9	277.17	168.78
Mask2Former[66]	1024^2	41.94	192.62	263.78
MPCNet[67]	1024^2	35.3	524.95	3052.76
Xnet[68]	1024^2	326.18	5343.39	1269.87
RSMamba[38]	1024^2	40.73	149.11	2499.72
VMFormer[69]	1024^2	27.4	164.35	327.91
Samba[40]	1024^2	97.83	294.91	4768.26
CPSSNet(Ours)	1024^2	31.73	**110.65**	147.54

effectively handles such features by establishing long-range dependencies in both spatial and channel dimensions.

Fig. 19 presents the segmentation results of various methods on the LoveDA dataset, as outlined in Table VII. From the second and third rows of Fig. 19, it can be observed that CPSSNet excels in segmenting large, continuous structures such as agricultural and barren areas with similar colors. This highlights the strong spatial long-range modeling capability of the SMB module in CPSSNet. In contrast, other Transformer and CNN networks often struggle to effectively differentiate subtle variations in such similar regions. In addition, as seen in the first, fourth, and fifth rows of Fig. 19, CPSSNet's cross-channel dynamic modeling, enabled by the CMB module, significantly enhances its ability to segment long, continuous objects. This advantage becomes even more pronounced when dealing with complex scenes and high-similarity features, further demonstrating the effectiveness of CPSSNet.

Observing Table VII, it can be seen that CPSSNet does not achieve the highest metrics in some categories such as barren and forest and performs poorly in segmenting building and water categories. This is due to the presence of features with different scales and sizes in these categories, a characteristic of the LoveDA dataset. Since we did not incorporate a dedicated multiscale processing module and only performed simple multiscale feature fusion at the initial stage, CPSSNet struggles with categories exhibiting significant multiscale variations. Fig. 20 clearly illustrates this issue, showing poor segmentation performance for building and forest categories with evident multiscale variations. In addition, CPSSNet does not have targeted edge processing, resulting in unsatisfactory edge segmentation performance across various categories, as shown in Fig. 20.

V. LIMITATIONS AND FUTURE PROSPECTS

Although CPSSNet demonstrates superior modeling capabilities at both spatial and channel levels, effectively addressing the color similarity issue in remote sensing image segmentation, it still has some limitations. First, CPSSNet does not specifically address multiscale characteristics, which are crucial in remote sensing image segmentation because these images typically contain objects of various scales and complex background information. Effectively capturing and utilizing these multiscale features can significantly enhance segmentation accuracy and model robustness. CPSSNet's performance tends to degrade when dealing with data with pronounced multiscale characteristics. Second, the framework lacks structures specifically designed for edge processing, leading to suboptimal edge segmentation performance. In future work, we plan to address edges specifically through SSM and integrate multiscale features into the network framework to develop a more generalizable segmentation model. In addition, we aim to focus on model compression to enhance the model's efficiency and practicality.

VI. CONCLUSION

This article designs a remote sensing segmentation network, CPSSNet, based on spatial SSMs and channel SSMs to address the challenge of color similarity in remote sensing segmentation. The network handles feature patching to establish long-range dependencies with linear complexity at the spatial level, utilizing state-space modeling at the spatial layer while also fully exploring spatial information through weighted channel-layer state-space modeling to achieve cross-channel stream interactions. Specifically, we first perform patching on the sampled features. At the spatial layer, the SMB module flattens the input features into sequences along four different directions and then models them using the SSM, effectively establishing long-range dependencies. Meanwhile, to achieve cross-channel information interaction, we propose a CMB. This module dynamically allocates weights by using the hidden state of the previous channel to adjust the subsequent channel weights. Given that the original SSM model cannot support dynamic weight adjustment, we introduce dynamic weight coefficients determined by the hidden state from the previous stage. Experimental results show that the SMB effectively establishes long-range dependencies quickly, while the CMB also provides efficient and rapid modeling of stream channel information for remote sensing segmentation. CPSSNet, composed of these two core components, SMB and CMB, demonstrates superior segmentation performance compared to mainstream networks in the remote sensing segmentation field. We hope that CPSSNet will provide valuable insights for future researchers in modeling at the channel level using SSM and addressing the color similarity problem in remote sensing segmentation.

REFERENCES

[1] R. Li, S. Zheng, C. Duan, L. Wang, and C. Zhang, "Land cover classification from remote sensing images based on multi-scale fully convolutional network," *Geo-Spatial Inf. Sci.*, vol. 25, no. 2, pp. 278–294, Jan. 2022.

[2] D. Marcos, M. Volpi, B. Kellenberger, and D. Tuia, "Land cover mapping at very high resolution with rotation equivariant CNNs: Towards small yet accurate models," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 96–107, Nov. 2018.

[3] J. Xing, R. Sieber, and T. Caelli, "A scale-invariant change detection method for land use/cover change research," *ISPRS J. Photogramm. Remote Sens.*, vol. 141, pp. 252–264, Jul. 2018.

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[4] I. de Gélis, S. Lefèvre, and T. Corpetti, "Siamese KPConv: 3D multiple change detection from raw point clouds using deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 197, pp. 274–291, Mar. 2023.

[5] A. Samie et al., "Examining the impacts of future land use/land cover changes on climate in Punjab province, pakistan: Implications for environmental sustainability and economic growth," *Environ. Sci. Pollut. Res.*, vol. 27, no. 20, pp. 25415–25433, Jul. 2020.

[6] F. Chen, H. Balzter, F. Zhou, P. Ren, and H. Zhou, "DGNet: Distribution guided efficient learning for oil spill image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4201317, doi: 10.1109/TGRS.2023.3240579.

[7] D. Griffiths and J. Boehm, "Improving public data for building segmentation from convolutional neural networks (CNNs) for fused airborne LiDAR and image data using active contours," *ISPRS J. Photogramm. Remote Sens.*, vol. 154, pp. 70–83, Aug. 2019.

[8] P. Shamsolmoali, M. Zareapoor, H. Zhou, R. Wang, and J. Yang, "Road segmentation for remote sensing images using adversarial spatial pyramid networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 4673–4688, Jun. 2021.

[9] J. Li, X. Feng, and Z. Hua, "Low-light image enhancement via progressive-recursive network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 11, pp. 4227–4240, Nov. 2021.

[10] Z. Huang, J. Li, Z. Hua, and L. Fan, "Underwater image enhancement via adaptive group attention-based multiscale cascade transformer," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–18, 2022.

[11] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[12] X. Liu et al., "Deep multiview union learning network for multisource image classification," *IEEE Trans. Cybern.*, vol. 52, no. 6, pp. 4534–4546, Jun. 2022.

[13] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.

[14] X. Wang, Z. Hu, S. Shi, M. Hou, L. Xu, and X. Zhang, "A deep learning method for optimizing semantic segmentation accuracy of remote sensing images based on improved UNet," *Sci. Rep.*, vol. 13, no. 1, p. 7600, May 2023.

[15] Y. Liu, H. Li, C. Hu, S. Luo, Y. Luo, and C. Wen Chen, "Learning to aggregate multi-scale context for instance segmentation in remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 595–609, Jan. 2025.

[16] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[17] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2018, pp. 7132–7141.

[18] H. Sun et al., "Unsupervised bidirectional contrastive reconstruction and adaptive fine-grained channel attention networks for image dehazing," *Neural Netw.*, vol. 176, Aug. 2024, Art. no. 106314.

[19] M. Zhu, W. Min, J. Han, Q. Han, and S. Cui, "Improved channel attention methods via hierarchical pooling and reducing information loss," *Pattern Recognit.*, vol. 148, Apr. 2024, Art. no. 110148.

[20] C. Fu, B. Du, and L. Zhang, "ReSC-Net: Hyperspectral image classification based on attention-enhanced residual module and spatial-channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518615.

[21] J. Park, S. Woo, J.-Y. Lee, and I. So Kweon, "BAM: Bottleneck attention module," 2018, *arXiv:1807.06514*.

[22] D. C. Bui, T. V. Le, and B. H. Ngo, "C^{2}T-Net: Channel-aware cross-fused transformer-style networks for pedestrian attribute recognition," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops (WACVW)*, Jan. 2024, pp. 351–358.

[23] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7262–7272.

[24] T. Xiao, Y. Liu, Y. Huang, M. Li, and G. Yang, "Enhancing multiscale representations with transformer for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605116.

[25] H. Cao et al., "Swin-UNet: UNet-like pure transformer for medical image segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 205–218.

[26] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[27] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[28] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[29] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117, doi: 10.1109/TGRS.2024.3356074.

[30] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis, "Efficient streaming language models with attention sinks," 2023, *arXiv:2309.17453*.

[31] P. Kacham, V. Mirrokni, and P. Zhong, "PolySketchFormer: Fast transformers via sketching polynomial kernels," 2023, *arXiv:2310.01655*.

[32] J. Ding et al., "LongNet: Scaling transformers to 1,000,000,000 tokens," 2023, *arXiv:2307.02486*.

[33] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[34] R. E. Kalman, "A new approach to linear filtering and prediction problems," *J. Basic Eng.*, vol. 82, no. 1, pp. 35–45, Mar. 1960.

[35] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[36] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.

[37] L. Wang, D. Li, S. Dong, X. Meng, X. Zhang, and D. Hong, "PyramidMamba: Rethinking pyramid feature fusion with selective space state model for semantic segmentation of remote sensing imagery," 2024, *arXiv:2406.10828*.

[38] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-mamba for large remote sensing image dense prediction," 2024, *arXiv:2404.02668*.

[39] X. Ma, X. Zhang, and M.-O. Pun, "RS3Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[40] Q. Zhu et al., "Samba: Semantic segmentation of remotely sensed images with state space model," *Heliyon*, vol. 10, no. 19, Oct. 2024, Art. no. e38495.

[41] R. Li, S. Zheng, C. Duan, J. Su, and C. Zhang, "Multistage attention ResU-net for semantic segmentation of fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[42] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2020.

[43] K.-H. Liu and B.-Y. Lin, "MSCSA-Net: Multi-scale channel spatial attention network for semantic segmentation of remote sensing images," *Appl. Sci.*, vol. 13, no. 17, p. 9491, Aug. 2023.

[44] X. Su, J. Li, and Z. Hua, "Transformer-based regression network for pansharpening remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5407423, doi: 10.1109/TGRS.2022.3152425.

[45] G. Yuan, J. Song, and J. Li, "IF-USOD: Multimodal information fusion interactive feature enhancement architecture for underwater salient object detection," *Inf. Fusion*, vol. 117, May 2025, Art. no. 102806.

[46] Y. Li et al., "EfficientFormer: Vision transformers at MobileNet speed," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, Jan. 2022, pp. 12934–12949.

[47] X. Liu, H. Peng, N. Zheng, Y. Yang, H. Hu, and Y. Yuan, "EfficientViT: Memory efficient vision transformer with cascaded group attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14420–14430.

[48] S. Mehta and M. Rastegari, "MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer," 2021, *arXiv:2110.02178*.

[49] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[50] Y. Fu, J. Liu, and J. Shi, "TSCA-Net: Transformer based spatial-channel attention segmentation network for medical images," *Comput. Biol. Med.*, vol. 170, Mar. 2024, Art. no. 107938.

[51] A. Gu, T. Dao, S. Ermon, A. Rudra, and C. Ré, "HiPPO: Recurrent memory with optimal polynomial projections," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, Jan. 2020, pp. 1474–1487.

[52] A. Gu et al., "Combining recurrent, convolutional, and continuous-time models with linear state space layers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 572–585.

[53] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

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[54] R. Hasani, M. Lechner, T.-H. Wang, M. Chahine, A. Amini, and D. Rus, "Liquid structural state-space models," 2022, *arXiv:2209.12951*.
[55] J. T. H. Smith, A. Warrington, and S. W. Linderman, "Simplified state space layers for sequence modeling," 2022, *arXiv:2208.04933*.
[56] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," 2024, *arXiv:2404.03425*.
[57] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Remote sensing image change detection with mamba," 2024, *arXiv:2406.04207*.
[58] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 11976–11986.
[59] G. Blelloch, "Prefix sums and their applications," School Comput. Sci., Carnegie Mellon Univ., Pittsburgh, PA, USA, Tech. Rep. CU-CS-90-190, 1990.
[60] E. Martin and C. Cundy, "Parallelizing linear recurrent neural nets over sequence length," 2017, *arXiv:1709.04057*.
[61] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.
[62] K. Chen, Z. Zou, and Z. Shi, "Building extraction from remote sensing images with sparse token transformers," *Remote Sens.*, vol. 13, no. 21, p. 4441, Nov. 2021.
[63] R. Li, L. Wang, C. Zhang, C. Duan, and S. Zheng, "A^2-FPN for semantic segmentation of fine-resolution remotely sensed images," *Int. J. Remote Sens.*, vol. 43, no. 3, pp. 1131–1155, Feb. 2022.
[64] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.
[65] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[66] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1290–1299.
[67] Q. Wang, X. Luo, J. Feng, G. Zhang, X. Jia, and J. Yin, "Multi-scale prototype contrast network for high-resolution aerial imagery semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615114.
[68] Y. Zhou, J. Huang, C. Wang, L. Song, and G. Yang, "XNet: Wavelet-based low and high frequency fusion networks for fully-and semi-supervised semantic segmentation of biomedical images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jun. 2023, pp. 21085–21096.
[69] H. Yan, M. Wu, and C. Zhang, "Multi-scale representations by varying window attention for semantic segmentation," 2024, *arXiv:2404.16573*.



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